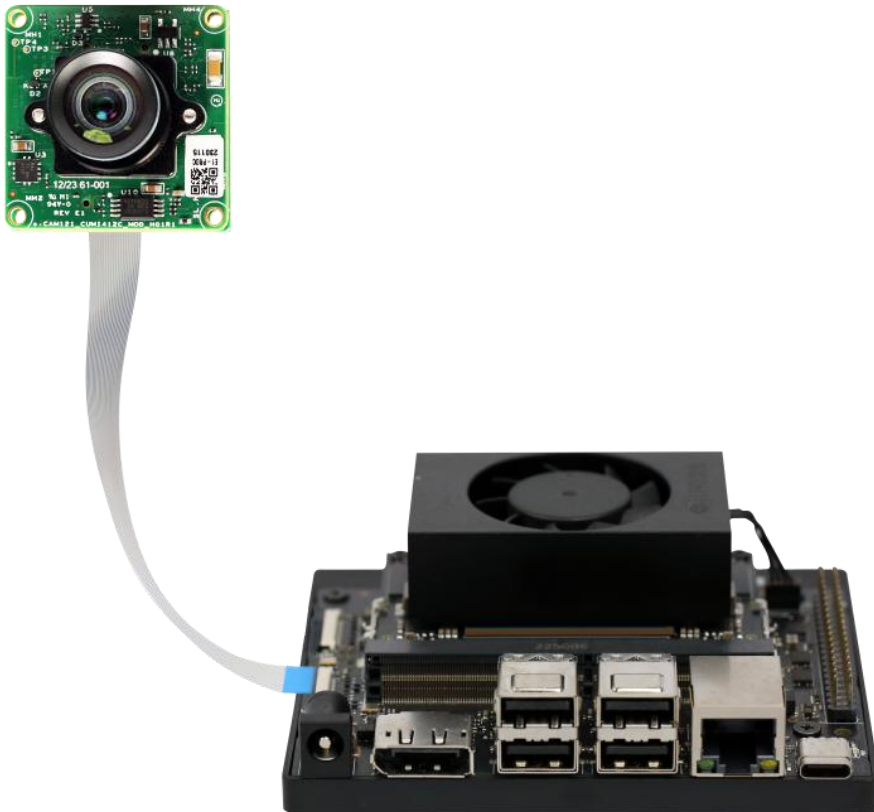


e-CAM121_CUONX

Developer Guide



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e-con Systems

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Disclaimer

e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

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Introduction to e-CAM121_CUONX

e-CAM121_CUONX is a new Mobile Industry Processor Interface (MIPI) camera from e-con Systems, a company with over two decades of experience in designing, developing, and manufacturing OEM cameras. The e-CAM121_CUONX consists of IMX412-AACK-D camera module, which are connected with a NVIDIA® Jetson AGX Orin™ development kit in 4-lane mode. The prebuild driver for this camera along with the camera board is provided by e-con Systems.

e-CAM121_CUONX is a variant of 12.3 MP color camera and is provided with S-mount (also known as M12 board lens) lens holder. The S-mount is one of the most commonly used small form-factor lens mounts for board camera. The MIPI camera contains a 1/2.3" IMX412-AACK-D CMOS image sensor from SONY® and is interfaced to a J21 connector of the Orin Nano™ /Orin™ NX development kit using an ACC-ONX-MI-WTB-ADP-H01 adaptor board.

The NVIDIA® Jetson Orin Nano™ /Orin™ NX development kit is a small, powerful computer for embedded applications and Artificial Intelligence (AI) Internet of Things (IoT) and is supported by NVIDIA® complete development toolchain. It is pre-flashed with a Linux environment, which includes support for many common APIs. The development kit supports a single camera for 4-lane configuration.

e-con Systems also provides the sample applications that demonstrates the features of this camera. However, this camera can utilize any Video for Linux version 2 (V4L2) application.

The commands and output messages in this manual are represented by different colors as shown in the below table.

Table 1: Notation of Colors

Color	Notation
Blue	Commands running in Host PC
Red	Commands running in Development Kit
Green	Output message in Terminal

This document explains how to setup the NVIDIA® Jetson Orin Nano™ /Orin™ NX for using e-CAM121_CUONX camera.

Software Requirements

The software requirements are as follows:

- Cross compiler toolchain.

- Linux for Tegra (L4T) release package and sample root filesystems (rootfs).

Prerequisites for Jetson Orin Nano

This section describes the requirements to use e-CAM121_CUONX on NVIDIA® Jetson Orin Nano™ development kit.

The prerequisites are as follows:

- Host PC which runs Ubuntu 20.04 (64-bit) or Ubuntu 18.04 (64-bit).
- Download and install lbzip2 package.
- NVIDIA® provided L4T release and corresponding sample rootfs for NVIDIA® Jetson Orin Nano™ development kit.
- The USB Type-C cable to plug into the recovery port of the Jetson Orin Nano™ development kit.
- Power cable (19V) to power the Jetson Orin Nano™ development kit.
- 64GB USB boot drive [SD Card/Pendrive] must be connected to the respective slot.
- A kernel image and device tree blob (DTB file). The release package contains the kernel binary (image) and the DTB files, which you can download and rebuild from source.

Please refer to the e-CAM121_CUONX_Release_Package_Manifest_Rev_<ver>.pdf to know the contents of release package and their description.

Setting Up Environment

The steps to set up the environments are as follows:

1. Run the following commands to setup the required environment variables.

```
mkdir top_dir/kernel_out -p
mkdir top_dir/kernel_sources
export TOP_DIR=<absolute path to>/top_dir
export RELEASE_PACK_DIR=$TOP_DIR/e-
CAM121_CUONX_JETSON_ONX_ONANO<L4T_version>_<release_
date>_<release_version>
export L4T_DIR=$TOP_DIR/Linux_for_Tegra
export LDK_ROOTFS_DIR=$TOP_DIR/Linux_for_Tegra/rootfs
export ARCH=arm64
export CROSS_COMPILE=aarch64-buildroot-linux-gnu-
export CROSS32CC=arm-linux-gnueabi-gcc
export TEGRA_KERNEL_OUT=$TOP_DIR/kernel_out
```

```
export KERNEL_PATH=$TOP_DIR/kernel_out
export
NVIDIA_SRC=$TOP_DIR/kernel_sources/Linux_for_Tegra/source/p
ublic
export DISPLAY_SOURCE_DIR=$NVIDIA_SRC/NVIDIA-kernel-
module-source-TempVersion
```

2. Run the following command to copy the e-con Systems release package tar file to the staging directory.

```
mv <location of>/e-
CAM121_CUONX_JETSON_ONX_ONANO_<L4T_version>_<release
_date>_<release_version>.tar.gz $TOP_DIR
```

Note: The above steps must be performed in a single terminal so that the environment variables are preserved.

Downloading Requirements

For building the kernel, a cross compiler toolchain and other tools necessary for compiling are required. You can use the default cross compiler toolchain and other tools provided in Ubuntu repositories. The necessary tools can be installed by running the command on the host Linux PC.

The steps to download the requirements for building the kernel are as follows:

1. Download the required toolchain from NVIDIA® website using <https://developer.nvidia.com/embedded/downloads> link.

The steps to download the required toolchain and export PATH environment are as follows:

- a. Download the required toolchain from NVIDIA® website as listed in the below table.

Table 2: GCC Toolchain Package

Title	Version	Download link
GCC Tool Chain for 64-bit BSP	9.3.0	https://developer.nvidia.com/embedded/jetson-linux/bootlin-toolchain-gcc-93

- b. Run the following command to extract the package in host PC.

```
mkdir aarch64-buildroot-linux-gnu
tar -xf ~/Downloads/aarch64--glibc--stable-final.tar.gz -C
aarch64-buildroot-linux-gnu
```


- c. Run the following command to export PATH environment for building kernel source.

```
export PATH=<Tool_chain_extract_path>/aarch64-buildroot-linux-gnu/bin:$PATH
```

2. Run the following commands to download the pre-requisite packages.

```
sudo apt-get update
sudo apt-get install lbzip2 qemu-user-static python build-essential bc flex openssl libssl-dev
```

3. Download the required L4T release package and sample root filesystem from NVIDIA® website using the provided links.

The steps to download the packages and copy the downloaded file are as follows:

- a. Download the packages from the NVIDIA® website as listed in the below table.

Table 3: Packages for Jetson Orin Nano

Title	Version	Download Link
L4T Jetson Orin Nano™ R35.4.1 Jetson™ Driver Package	35.4.1	https://developer.nvidia.com/downloads/embedded/l4t/r35_release_v4.1/release/jetson_linux_r35.4.1_aarch64.tbz2/
L4T Jetson Orin Nano™ R35.4.1 Sample Root File system	35.4.1	https://developer.nvidia.com/downloads/embedded/l4t/r35_release_v4.1/release/tegra_linux_sample-root-filesystem_r35.4.1_aarch64.tbz2/

- b. Run the following commands to copy the downloaded file to staging directory.

```
cp $HOME/Downloads/Jetson_Linux_R35.4.1_aarch64.tbz2 $TOP_DIR
cp $HOME/Downloads/Tegra_Linux_Sample-Root-Filesystem_R35.4.1_aarch64.tbz2 $TOP_DIR
```

Extracting and Preparing L4T

The steps for extracting and preparing L4T are as follows:

Note: The following steps must be performed in the host PC.

1. Run the following commands to extract the downloaded L4T release package to navigate a folder with the name Linux_for_Tegra.

```
cd $TOP_DIR  
sudo tar -I lbzip2 -xpf Jetson_Linux_R35.4.1_aarch64.tbz2
```

Note: The folder contains the necessary tools and binaries for modifying the NVIDIA® Jetson Orin Nano™ development kit.

2. Run the following commands to extract the sample file system to the rootfs directory which is present inside the Linux_for_Tegra directory.

```
cd $LDK_ROOTFS_DIR  
sudo tar -I lbzip2 -xpf $TOP_DIR/Tegra_Linux_Sample-Root-  
Filesystem_R35.4.1_aarch64.tbz2
```

3. Run the following commands to set the package to be ready to flash binaries.

```
cd $L4T_DIR  
sudo ./apply_binaries.sh  
sudo ./tools/l4t_flash_prerequisites.sh
```

Extracting Release Package

Run the following commands to extract the e-CAM121_CUONX release package.

```
cd $TOP_DIR  
tar -xaf e-  
CAM121_CUONX_JETSON_ONX_ONANO_<L4T_version>_<release_d  
ate>_<release_version>.tar.gz
```

To know more about the release package, please refer to the e-CAM121_CUONX_Release_Package_Manifest_Rev_<ver>.pdf.

Installation Procedure for Jetson Orin Nano

This section describes the steps for building and installing the kernel. You can choose to use own customized kernel.

Building from Source

You can use the patch file provided by e-con Systems to build your own kernel image binary and modules with support to use e-CAM121_CUONX camera on the NVIDIA® Jetson Orin Nano™ development kit.

Downloading and Configuring the Kernel

This section describes how you can download and configure the kernel for Jetson Orin Nano™ development kit.

Download the kernel source code for L4T from the NVIDIA® website using <https://developer.nvidia.com/embedded/downloads> link.

The steps to download and configure the kernel for Jetson Orin Nano™ development kit are as follows:

1. Download the packages from the NVIDIA® website as listed in the below table.

Table 4: Packages for Jetson Orin Nano

Title	Version	Download Link
L4T Jetson Orin Nano™ R35.4.1 Sources	35.4.1	https://developer.nvidia.com/downloads/embedded/l4t/r35_release_v4.1/sources/public_sources.tbz2/

2. Run the following command to copy the downloaded file to staging directory.

```
cp $HOME/Downloads/public_sources.tbz2
$TOP_DIR/kernel_sources
```

3. Run the following commands to extract the downloaded kernel source code to any path on the host Linux PC.

```
cd $TOP_DIR/kernel_sources
tar -I lbzip2 -xpf public_sources.tbz2
cd $NVIDIA_SRC
```

4. Run the following commands to extract the kernel source code.

```
tar -I lbzip2 -xpf kernel_src.tbz2
tar -I lbzip2 -xpf nvidia_kernel_display_driver_source.tbz2
```

5. Run the following commands to make sure that all the three patch commands are applied properly in the kernel source.

```
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_kernel.patch -
-dry-run
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_dtb.patch --
dry-run
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_module.patch
--dry-run
```

6. Run the following commands to apply the three patch files to the kernel source code if there is no error from dry-run command.

```
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_kernel.patch
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_dtb.patch
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_module.patch
```

Building and Installing Kernel

Run the following commands to build and install the kernel image and modules to the NVIDIA® Jetson Orin Nano™ development kit.

```
cd kernel/kernel-5.10/
make O=$TEGRA_KERNEL_OUT tegra_ecam_isp_multicam_defconfig
make O=$TEGRA_KERNEL_OUT Image -j4
make O=$TEGRA_KERNEL_OUT modules -j4
make O=$TEGRA_KERNEL_OUT dtbs
sudo ARCH=arm64 make O=$TEGRA_KERNEL_OUT modules_install
INSTALL_MOD_PATH=$LDK_ROOTFS_DIR
cd $NVIDIA_SRC/e-CAM121_CUONX
make
sudo make -C $TEGRA_KERNEL_OUT M=$PWD
INSTALL_MOD_PATH=$LDK_ROOTFS_DIR modules_install
cd $DISPLAY_SOURCE_DIR
```

```
make -C $PWD modules -j $(nproc) TARGET_ARCH=aarch64
ARCH=arm64 CC="${CROSS_COMPILE}gcc"
LD="${CROSS_COMPILE}ld" AR="${CROSS_COMPILE}ar"
CXX="${CROSS_COMPILE}g++"
OBJCOPY="${CROSS_COMPILE}objcopy"
SYSOUT="$TEGRA_KERNEL_OUT"
SYSSRC="$NVIDIA_SRC/kernel/kernel-5.10"

sudo make M=$PWD/kernel-open -C $TEGRA_KERNEL_OUT
INSTALL_MOD_PATH=$LDK_ROOTFS_DIR modules_install
```

Run the following commands to copy the kernel and dtb file to Linux_for_Tegra (L4T_DIR) flashing path.

For Jetson Orin Nano™ development kit,

```
sudo cp $TEGRA_KERNEL_OUT/arch/arm64/boot/Image
$L4T_DIR/kernel/ -f

sudo cp
$TEGRA_KERNEL_OUT/arch/arm64/boot/dts/nvidia/tegra234-p3767-
0003-p3768-0000-a0-4lane-imx412.dtb
$L4T_DIR/kernel/dtb/tegra234-p3767-0003-p3768-0000-a0.dtb -f
```

Note: Even if the image is custom built, the kernel configuration must have module versioning support for the camera driver to work.

Modifying Rootfs

The steps to modify additional files in the rootfs for the proper functioning of the e-CAM121_CUONX camera on the Jetson Orin Nano™ development kit are as follows:

ISP Libraries for Jetson Orin Nano™,

1. Run the following command to copy the file.

Note: Untar the **e-**

CAM121_CUONX_L4T35.4.1 JP5.1.2 JETSON-ONX-ONANO_R02.tar.gz file present inside the release package and then run the below commands.

```
cd $RELEASE_PACK_DIR
tar -xaf e-
CAM121_CUONX_<L4T_version>_<JP_version>_JETSON-ONX-
ONANO_<release_version>.tar.gz

sudo cp $RELEASE_PACK_DIR/e-
CAM121_CUONX_<L4T_version>_<JP_version>_JETSON-ONX-
ONANO_<release_version>/ONX_ONANO/misc/camera_overrides
_jetson-onano.isp
$LDK_ROOTFS_DIR/var/nvidia/nvcam/settings/camera_overrides.i
sp -f
```

2. Run the following commands for changing the file mode and owner of the file.

```
sudo chmod 664  
$LDK_ROOTFS_DIR/var/nvidia/nvcam/settings/camera_overrides.i  
sp  
sudo chown root:root  
$LDK_ROOTFS_DIR/var/nvidia/nvcam/settings/camera_overrides.i  
sp
```

Note: To achieve the best image quality, the ISP has been tuned by e-con Systems in collaboration with NVIDIA®, specific to e-CAM121_CUONX and the ISP configuration file is **camera_overrides.isp** file.

3. Run the following commands for boosting the clock speed to maximum speed for changing the file mode of the other file and changing owner of the other file.

```
sudo cp $RELEASE_PACK_DIR/e-  
CAM121_CUONX_L4T35.4.1_JP5.1.2_JETSON-ONX-  
ONANO_<release_version>/ONX_ONANO/misc/max-isp-vi-clks.sh  
$LDK_ROOTFS_DIR/home/max-isp-vi-clks.sh -f  
sudo chmod +x $LDK_ROOTFS_DIR/home/max-isp-vi-clks.sh
```

Other files,

```
sudo cp $RELEASE_PACK_DIR/e-  
CAM121_CUONX_L4T35.4.1_JP5.1.2_JETSON-ONX-  
ONANO_<release_version>/ONX_ONANO/misc/modules  
$LDK_ROOTFS_DIR/etc/modules -f
```

Flashing Jetson Orin Nano Development Kit

The steps to flash the Jetson Orin Nano™ development kit are as follows:

1. Power OFF the kit.
2. Connect a jumper across FC-REC and GND to enter recovery mode.
3. Connect the Type-C USB cable to the Jetson Orin Nano™ development kit and host PC.

The top view of the Jetson Orin Nano™ development kit is shown below.

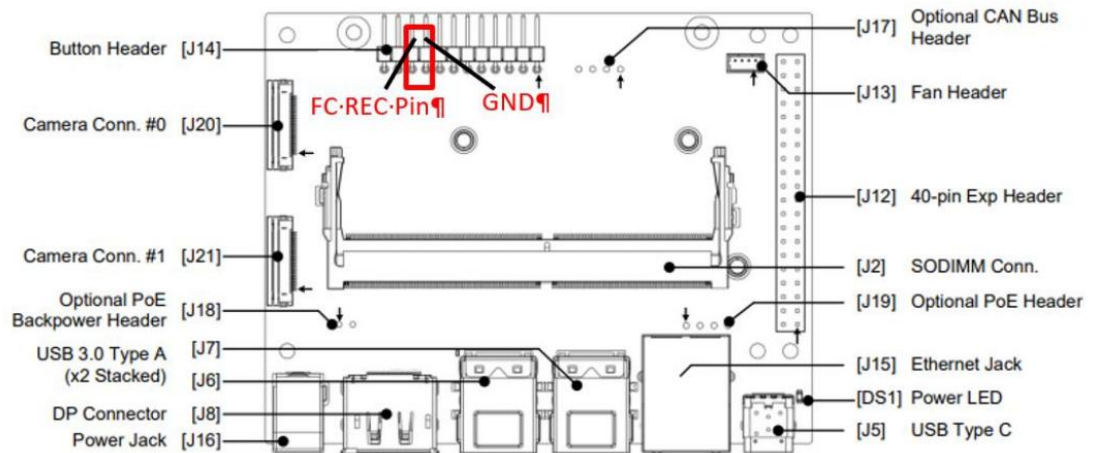


Figure 1: Top View of Jetson Nano Development Kit

4. Power ON the Jetson Orin Nano™ development kit.

If the Jetson Orin Nano™ development kit is successfully changed to recovery mode, the Jetson Orin Nano™ development kit will be enumerated as an USB device to the host PC.

5. Run the following command to verify whether the board is in recovery mode.

```
lsusb
```

The output message appears as shown below.

```
Bus <bbb> Device <ddd>: ID 0955:7523 NVidia Corp.
```

Where:

<bbb> = is any three-digit number.

<ddd> = is any three-digit number.

6. Run the following commands to flash the Jetson Orin Nano™ development kit from your host PC.

```
cd $L4T_DIR
```

1. Flash to NVMe

```
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device  
nvme0n1p1 -c tools/kernel_flash/flash_l4t_external.xml -p "-c  
bootloader/t186ref/cfg/flash_t234_qspi.xml" --showlogs --network  
usb0 jetson-orin-nano-devkit internal
```

2. Flash to the USB SSD

```
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device sda1  
-c tools/kernel_flash/flash_l4t_external.xml -p "-c  
bootloader/t186ref/cfg/flash_t234_qspi.xml" --showlogs --network  
usb0 jetson-orin-nano-devkit internal
```

3. Flash to the SD

```
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device  
mmcblk1p1 -c tools/kernel_flash/flash_l4t_external.xml -p "-c  
bootloader/t186ref/cfg/flash_t234_qspi.xml" --showlogs --network  
usb0 jetson-orin-nano-devkit internal
```

Note: If you are flashing with a USB flash drive (Pendrive or SD card) as storage, the USB drive must be at least 64GB.

Note: Now, the entire USB drive storage/NVMe on the Jetson Orin Nano™ development kit will be erased. Flashing will take about 10-30 minutes to complete depending on the host PC configuration.

7. Configure the Jetson Orin Nano™ development kit to a monitor and a keyboard.
8. Reboot the Jetson Orin Nano™ development kit to complete the OS configuration, after flashing.

Prerequisites for Jetson Orin NX

This section describes the requirements to use e-CAM121_CUONX on NVIDIA® Jetson Orin Nano™ development kit with a Jetson Orin™ NX SOM.

The prerequisites are as follows:

- Host PC which runs Ubuntu 20.04 (64-bit) or Ubuntu 18.04 (64-bit).
- Download and install lbzip2 package.
- NVIDIA® provided L4T release and corresponding sample rootfs for NVIDIA® Jetson Orin™ NX development kit.
- The USB Type-C cable to plug into the recovery port of the Jetson Orin™ NX development kit.
- Power cable (19V) to power the Jetson Orin™ NX development kit.
- 64GB USB boot drive [SD Card/Pendrive] must be connected to the respective slot.
- A kernel image and device tree blob (DTB file). The release package contains the kernel binary (image) and the DTB files, which you can download and rebuild from source.

Please refer to the e-CAM121_CUONX_Release_Package_Manifest_Rev_<ver>.pdf to know the contents of release package and their description.

Setting Up Environment

The steps to set up the environments are as follows:

1. Run the following commands to setup the required environment variables.

```
mkdir top_dir/kernel_out -p
mkdir top_dir/kernel_sources
export TOP_DIR=<absolute path to>/top_dir
export RELEASE_PACK_DIR=$TOP_DIR/e-
CAM121_CUONX_JETSON_ONX_ONANO<L4T_version>_<release_
date>_<release_version>
export L4T_DIR=$TOP_DIR/Linux_for_Tegra
export LDK_ROOTFS_DIR=$TOP_DIR/Linux_for_Tegra/rootfs
export ARCH=arm64
export CROSS_COMPILE=aarch64-buildroot-linux-gnu-
export CROSS32CC=arm-linux-gnueabi-hf-gcc
```

```
export TEGRA_KERNEL_OUT=$TOP_DIR/kernel_out
export KERNEL_PATH=$TOP_DIR/kernel_out
export
NVIDIA_SRC=$TOP_DIR/kernel_sources/Linux_for_Tegra/source/p
ublic
export DISPLAY_SOURCE_DIR=$NVIDIA_SRC/NVIDIA-kernel-
module-source-TempVersion
```

2. Run the following command to copy the e-con Systems release package tar file to the staging directory.

```
mv <location of>/e-
CAM121_CUONX_JETSON_ONX_ONANO_<L4T_version>_<release
_date>_<release_version>.tar.gz $TOP_DIR
```

Note: The above steps must be performed in a single terminal so that the environment variables are preserved.

Downloading Requirements

For building the kernel, a cross compiler toolchain and other tools necessary for compiling are required. You can use the default cross compiler toolchain and other tools provided in Ubuntu repositories. The necessary tools can be installed by running the command on the host Linux PC.

The steps to download the requirements for building the kernel are as follows:

1. Download the required toolchain from NVIDIA® website using <https://developer.nvidia.com/embedded/downloads> link.

The steps to download the required toolchain and export PATH environment are as follows:

- a. Download the required toolchain from NVIDIA® website as listed in the below table.

Table 5: GCC Toolchain Package

Title	Version	Download link
GCC Tool Chain for 64-bit BSP	9.3.0	https://developer.nvidia.com/embedded/jetson-linux/bootlin-toolchain-gcc-93

- b. Run the following commands to extract the package in host PC.

```
mkdir aarch64-buildroot-linux-gnu
tar -xf ~/Downloads/aarch64--glibc--stable-final.tar.gz -C
aarch64-buildroot-linux-gnu
```

- c. Run the following command to export PATH environment for building the kernel source.

```
export PATH=<Tool_chain_extract_path>/aarch64-buildroot-linux-gnu/bin:$PATH
```

2. Run the following commands to download the pre-requisite packages.

```
sudo apt-get update
sudo apt-get install lbzip2 qemu-user-static python build-essential bc flex openssl libssl-dev
```

3. Download the required L4T release package and sample root filesystem from NVIDIA® website using the provided links.

The steps to download the packages and copy the downloaded file are as follows:

- a. Download the packages from the NVIDIA® website as listed in the below table.

Table 6: Packages for Jetson Orin NX

Title	Version	Download Link
L4T Jetson Orin™ NX R35.4.1 Jetson™ Driver Package	35.4.1	https://developer.nvidia.com/downloads/embedded/l4t/r35_release_v4.1/release/jetson_linux_r35.4.1_aarch64.tbz2/
L4T Jetson Orin™ NX R35.4.1 Sample Root File system	35.4.1	https://developer.nvidia.com/downloads/embedded/l4t/r35_release_v4.1/release/tegra_linux_sample-root-filesystem_r35.4.1_aarch64.tbz2/

- b. Run the following commands to copy the downloaded file to staging directory.

```
cp $HOME/Downloads/Jetson_Linux_R35.4.1_aarch64.tbz2 $TOP_DIR
cp $HOME/Downloads/Tegra_Linux_Sample-Root-Filesystem_R35.4.1_aarch64.tbz2 $TOP_DIR
```

Extracting and Preparing L4T

The steps to extract and prepare L4T are as follows:

Note: The following steps must be performed in the host PC.

1. Run the following commands to extract the downloaded L4T release package to navigate a folder with the name Linux_for_Tegra.

```
cd $TOP_DIR  
sudo tar -I lbzip2 -xpf Jetson_Linux_R35.4.1_aarch64.tbz2
```

Note: The folder contains the necessary tools and binaries for modifying the NVIDIA® Jetson Orin™ NX development kit.

2. Run the following commands to extract the sample file system to the rootfs directory which is present inside the Linux_for_Tegra directory.

```
cd $LDK_ROOTFS_DIR  
sudo tar -I lbzip2 -xpf $TOP_DIR/Tegra_Linux_Sample-Root-  
Filesystem_R35.4.1_aarch64.tbz2
```

3. Run the following commands to set the package to be ready to flash binaries.

```
cd $L4T_DIR  
sudo ./apply_binaries.sh  
sudo ./tools/l4t_flash_prerequisites.sh
```

Extracting Release Package

Run the following commands to extract the e-CAM121_CUONX release package.

```
cd $TOP_DIR  
tar -xaf e-  
CAM121_CUONX_JETSON_ONX_ONANO_<L4T_version>_<release_d  
ate>_<release_version>.tar.gz
```

To know more about the release package, please refer to the e-CAM121_CUONX_Release_Package_Manifest_Rev_<ver>.pdf.

Installation Procedure for Jetson Orin NX

This section describes the steps for building and installing the kernel. You can choose to use own customized kernel.

Building from Source

You can use the patch file provided by e-con Systems to build your own kernel image binary and modules with support to use e-CAM121_CUONX camera on the NVIDIA® Jetson Orin Nano™ development kit with Jetson Orin™ NX SOM.

Downloading and Configuring the Kernel

This section describes how you can download and configure the kernel for Orin™ NX development kit .

Download the kernel source code for L4T from the NVIDIA® website using <https://developer.nvidia.com/embedded/downloads> link.

The steps to download and configure the kernel for Jetson Orin™ NX development kit are as follows:

1. Download the packages from the NVIDIA® website as listed in the below table.

Table 7: Packages for Jetson Orin NX

Title	Version	Download Link
L4T Jetson Orin™ NX R35.4.1 Sources	35.4.1	https://developer.nvidia.com/downloads/embedded/l4t/r35_release_v4.1/sources/public_sources.tbz2/

2. Run the following command to copy the downloaded file to staging directory.

```
cp $HOME/Downloads/public_sources.tbz2
$TOP_DIR/kernel_sources
```

3. Run the following commands to extract the downloaded kernel source code to any path on the host Linux PC.

```
cd $TOP_DIR/kernel_sources
tar -I lbzip2 -xpf public_sources.tbz2
cd $NVIDIA_SRC
```

4. Run the following commands to extract the kernel source code.

```
tar -I lbzip2 -xpf kernel_src.tbz2  
tar -I lbzip2 -xpf nvidia_kernel_display_driver_source.tbz2
```

5. Run the following commands to make sure that all the three patch commands are applied properly in the kernel source.

```
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-  
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_kernel.patch -  
-dry-run  
  
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-  
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_dtb.patch --  
dry-run  
  
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-  
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_module.patch  
--dry-run
```

6. Run the following commands to apply the three patch files to the kernel source code if there is no error from dry-run command.

```
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-  
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_kernel.patch  
  
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-  
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_dtb.patch  
  
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-  
CAM121_CUONX_JETSON_ONX_ONANO_L4T35.4.1_module.patch
```

Building and Installing Kernel

Run the following commands to build and install the kernel image and modules to the NVIDIA® Jetson Orin™ NX development kit.

```
cd kernel/kernel-5.10/  
make O=$TEGRA_KERNEL_OUT tegra_ecam_isp_multicam_defconfig  
make O=$TEGRA_KERNEL_OUT Image -j4  
make O=$TEGRA_KERNEL_OUT modules -j4  
make O=$TEGRA_KERNEL_OUT dtbs  
sudo ARCH=arm64 make O=$TEGRA_KERNEL_OUT modules_install  
INSTALL_MOD_PATH=$LDK_ROOTFS_DIR  
cd $NVIDIA_SRC/e-CAM121_CUONX  
make  
sudo make -C $TEGRA_KERNEL_OUT M=$PWD  
INSTALL_MOD_PATH=$LDK_ROOTFS_DIR modules_install
```

```
cd $DISPLAY_SOURCE_DIR
make -C $PWD modules -j $(nproc) TARGET_ARCH=aarch64
ARCH=arm64 CC="${CROSS_COMPILE}gcc"
LD="${CROSS_COMPILE}ld" AR="${CROSS_COMPILE}ar"
CXX="${CROSS_COMPILE}g++"
OBJCOPY="${CROSS_COMPILE}objcopy"
SYSOUT="$TEGRA_KERNEL_OUT"
SYSSRC="$NVIDIA_SRC/kernel/kernel-5.10"

sudo make M=$PWD/kernel-open -C $TEGRA_KERNEL_OUT
INSTALL_MOD_PATH=$LDK_ROOTFS_DIR modules_install
```

Run the following commands to copy the kernel and dtb file to Linux_for_Tegra (L4T_DIR) flashing path.

For Jetson Orin™ NX development kit,

```
sudo cp $TEGRA_KERNEL_OUT/arch/arm64/boot/Image
$L4T_DIR/kernel/ -f

sudo cp
$TEGRA_KERNEL_OUT/arch/arm64/boot/dts/nvidia/tegra234-p3767-
0000-p3768-0000-a0-4lane-imx412.dtb
$L4T_DIR/kernel/dtb/tegra234-p3767-0000-p3768-0000-a0.dtb -f
```

Note: Even if the image is custom built, the kernel configuration must have module versioning support for the camera driver to work.

Modifying Rootfs

The steps to modify additional files in the rootfs for the proper functioning of the e-CAM121_CUONX camera on the Jetson Orin™ NX development kit are as follows:

ISP Libraries for Jetson Orin™ NX,

1. Run the following command to copy the file.

Note: Untar the e-

CAM121_CUONX_L4T35.4.1_JP5.1.2_JETSON-ONX-ONANO_R02.tar.gz file present inside the release package and then run the below commands.

```
cd $RELEASE_PACK_DIR
tar -xaf e-
CAM121_CUONX_<L4T_version>_<JP_version>_JETSON-ONX-
ONANO_<release_version>.tar.gz

sudo cp $RELEASE_PACK_DIR/e-
CAM121_CUONX_<L4T_version>_<JP_version>_JETSON-ONX-
ONANO_<release_version>/ONX_ONANO/misc/camera_overrides
_jetson-onano.isp
$LDK_ROOTFS_DIR/var/nvidia/nvcam/settings/camera_overrides.i
sp -f
```

2. Run the following commands for changing the file mode and owner of the file.

```
sudo chmod 664  
$LDK_ROOTFS_DIR/var/nvidia/nvcam/settings/camera_overrides.i  
sp  
  
sudo chown root:root  
$LDK_ROOTFS_DIR/var/nvidia/nvcam/settings/camera_overrides.i  
sp
```

Note: To achieve the best image quality, the ISP has been tuned by e-con Systems in collaboration with NVIDIA®, specific to e-CAM121_CUONX and the ISP configuration file is **camera_overrides.isp** file.

3. Run the following commands for boosting the clock speed to maximum speed for changing the file mode of the other file and changing owner of the other file.

```
sudo cp $RELEASE_PACK_DIR/e-  
CAM121_CUONX_L4T35.4.1_JP5.1.2_JETSON-ONX-  
ONANO_<release_version>/ONX_ONANO/misc/max-isp-vi-clks.sh  
$LDK_ROOTFS_DIR/home/max-isp-vi-clks.sh -f  
  
sudo chmod +x $LDK_ROOTFS_DIR/home/max-isp-vi-clks.sh
```

Other files,

```
sudo cp $RELEASE_PACK_DIR/e-  
CAM121_CUONX_L4T35.4.1_JP5.1.2_JETSON-ONX-  
ONANO_<release_version>/ONX_ONANO/misc/modules  
$LDK_ROOTFS_DIR/etc/modules -f
```

Flashing Jetson Orin NX SOM + Jetson Orin Nano Development Kit

The steps to flash the Jetson Orin™ NX are as follows:

1. Power OFF the kit.
2. Connect the Orin™ NX SOM to the Jetson Orin Nano™ development kit.
3. Connect a jumper across FC-REC and GND to enter recovery mode.
4. Connect the Type-C USB cable to the Jetson Orin Nano™ development kit and host PC.

The top view of the Jetson Orin Nano™ development kit is shown below.

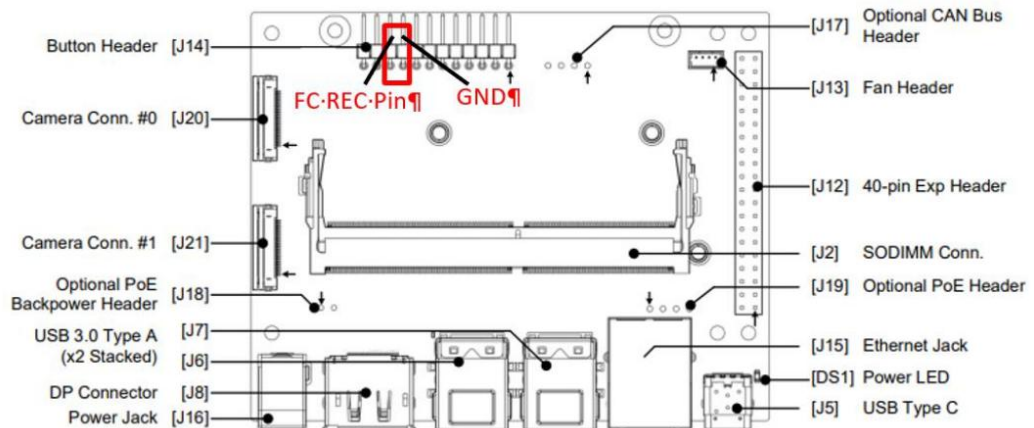


Figure 2: Top View of Jetson Nano Development Kit

5. Power ON the Jetson Orin Nano™ development kit.

If the Jetson Orin Nano™ development kit is successfully changed to recovery mode, the Jetson Orin™ NX SOM will be enumerated as an USB device to the host PC.

6. Run the following command to verify whether the board is in recovery mode.

```
lsusb
```

The output message appears as shown below.

```
Bus <bbb> Device <ddd>: ID 0955:7323 NVidia Corp.
```

Where:

<bbb> = is any three-digit number.

<ddd> = is any three-digit number.

7. Run the following commands to flash the Jetson Orin™ NX SOM + Jetson Orin Nano™ development kit from your host PC.

```
cd $L4T_DIR
```

1.Flash to NVMe

```
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device  
nvme0n1p1 -c tools/kernel_flash/flash_l4t_external.xml -p "-c  
bootloader/t186ref/cfg/flash_t234_qspi.xml" --showlogs --network  
usb0 jetson-orin-nano-devkit internal
```

4. Flash to the USB SSD

```
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device sda1  
-c tools/kernel_flash/flash_l4t_external.xml -p "-c
```

```
bootloader/t186ref/cfg/flash_t234_qspi.xml" --showlogs --network  
usb0 jetson-orin-nano-devkit internal
```

Note: If you are flashing with a USB flash drive (Pendrive or SD card) as storage, the USB drive must be at least 64GB.

Note: Now, the entire USB drive storage/NVMe on the Jetson Orin Nano™ development kit will be erased. Flashing will take about 10-30 minutes to complete depending on the host PC configuration.

8. Configure the Jetson Orin Nano™ development kit to a monitor and a keyboard.
9. Reboot the Jetson Orin Nano™ development kit to complete the OS configuration, after flashing.

Loading Drivers

This section describes how to load the drivers, install the sample application, and use the sample application with e-CAM121_CUONX.

The module drivers for e-CAM121_CUONX will be loaded automatically in the NVIDIA® Jetson Orin Nano™/Orin™ NX development kit during booting.

The steps to load the drivers are as follows:

1. Run the following command to check whether the camera connected are initialized.

```
dmesg | grep "Detected eimx412 sensor"
```

The output message appears as shown below.

```
Detected eimx412 sensor
```

The output message indicates that camera connected are initialized properly.

2. Run the following command to check the presence of video node.

```
ls /dev/video*
```

The output message for camera setup appears as shown below.

```
/dev/video0
```

Note: Only one camera node will be created for Jetson Orin Nano™/Orin™ NX development kit. Since only the CAM1 port on the Jetson Orin Nano™/Orin™ NX development kit is supported for the 4-lane configuration.

The video node reflects the connected camera to the Jetson Orin Nano™/Orin™ NX development kit. This video nodes can be utilized by any V4L2 application for viewing the camera preview.

Using Sample Applications with e-CAM121_CUONX

To use the eCAM_argus_camera application with e-CAM121_CUONX, please refer to the e-CAM121_CUONX_eCAM_Argus_App_User_Manual_Rev_<ver>.pdf.

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

Unloading the camera drivers causes a kernel crash.

This is a known issue. Please write to techsupport@e-consystems.com to get immediate support on other issues.

- 1. I have flashed L4T 35.4.1 in Jetson Orin Nano™/Orin™ NX development kit already. What are the steps to install the binaries?**

Refer e-
CAM121_CUONX_Getting_Started_Manual_Rev_<ver>.pdf to upgrade the latest binaries.

- 2. Is the provided camera driver binary .ko file compatible with all L4T version?**

No, it is not compatible with all L4T version. Please refer to *Downloading Requirements* section to know about the compatible L4T version.

Glossary

API: Application Programming Interface.

CMOS: Complementary Metal Oxide Semiconductor.

DTB: Device Tree Blob.

L4T: Linux for Tegra.

MIPI: Mobile Industry Processor Interface.

NVMe: Non-volatile Memory Express

Rootfs: Root Filesystem.

USB: Universal Serial Bus.

V4L2: Video for Linux version2.

Contact Us

If you need any support on e-CAM121_CUONX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	24-May-2023	Initial Draft	Product Marketing Team
1.1	01-July-2023	Support for both Orin Nano™ and Orin™ NX development kit	Product Marketing Team
1.2	27-August-2023	Supported Synchronous mode and 4K and 1080p resolution	Product Marketing Team